Reproducibility Report: Blair, Coppock and Moor (2020)

When to Worry about Sensitivity Bias: A Social Reference Theory and Evidence from 30 Years of List Experiments

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This repository holds the actively maintained replication code for Blair, Coppock and Moor (2020), together with the reproducibility report that documents what the original archive did and did not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi 10.1017/S1049096521000366) to a set of published archives.

Article	10.1017/S0003055420000374
Replication archive	10.7910/DVN/YUXHZT
Pre-analysis plans	none

The data are not redistributed here. The deposit is a single 1.2 MB zip at Harvard Dataverse, and that is the only copy this repository points at. `download_original.R` fetches it, verifies its checksum, and unpacks it into `original_extracted/`; `original_manifest.csv` records the Dataverse file identifier, the size in bytes, and two checksums: the MD5 of the bytes Dataverse serves for `?format=original`, which is what this code was written against, and the MD5 Dataverse publishes. Here the two agree. They do not always, so the script verifies against the served bytes and reports any disagreement. Either way the exact bytes this code ran on are pinned in version control even though the bytes themselves are not.

Repository layout. `maintained/` is the maintained rewrite: one script per published table or figure, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. `ground_truth/` ties every published number to the code that produces it: `published_claims.csv` is the extraction of what the article claims, `build_ground_truth.R` regenerates the comparison table from the pipeline's own output as the last step of `run_all.R`, and two scripts that run on their own record the other side of each comparison. `extract_archive_values.R` runs the deposited code in a scratch copy, which takes about a quarter of an hour; `extract_published_values.R` reads the supplementary materials PDF positionally, which needs a copy of the published appendix and so takes its location from an environment variable. `maintained/in_text_claims.R` prints every claimed number beside the sentence that makes it. `errata.qmd` at the repository root lists the sentences in the article that the data do not support and renders to `blair_coppock_moor_2020_errata.pdf`. `original/` and `original_extracted/` are created by the download script and are deliberately absent from the repository. This file is the reproducibility report, also available as a PDF in `report/`.

License. CC0 1.0 Universal, matching the terms of the deposit this repository maintains, so nothing in the chain is more restrictive than the archive itself. See `LICENSE`.

To reproduce. Clone or download the repository, open `blair_coppock_moor_2020.Rproj`, and run:

```
source("run_all.R")
```

That fetches the deposit, verifies it, produces every table and figure into `maintained/output/`, and rebuilds the ground truth table from those outputs. Required packages: tidyverse, rstan, tidybayes, deming, janitor, patchwork, DeclareDesign, bindata, rsample, here, and for this report knitr and kableExtra. Paths resolve through `here`, so nothing depends on the working directory. The full run takes about two minutes: a minute to recompile the Stan model

and fit the ten meta-analyses, another for the design simulation, and seconds for everything else. Individual scripts can be run on their own, subject to the three dependencies noted in `run_all.R`. A successful run overwrites `maintained/output/`, which is committed: **`git diff` on that folder is the reproduction check.** The CSV, PNG and RDS output comes back byte-identical; the 21 PDF figures always show as changed, because a PDF records the time it was written, so compare their PNG twins instead.

1 Summary

Two questions, answered before the detail.

1.1 Does the deposited archive run?

Not as deposited. The archive holds thirteen R files, two of which are helper files the others source. Of the eleven that are meant to be run, nine fail on a current R installation.

Three of the nine fail before reaching any R problem at all: **the deposit does not contain the output directories its own scripts write to.** `mode_appendix_table.R` and `table_of_data.R` stop at `cannot open file 'tables/...'`, and `plot_direct_v_list.R` stops at `Cannot find directory 'figures'`. Nothing in the archive creates either folder, so a reader who unpacks the zip and runs the scripts in the order the archive's README implies gets three failures that have nothing to do with the analysis and no hint that the fix is `mkdir`.

The core meta-analysis script, `meta_analysis.R`, loads `data/stan_model.RDS`, a Stan model compiled in 2020 under R 3.6 on another machine. rstan rejects it: "the compiled object from C++ code for this model is invalid." The archive ships `compile_stan_model.R` to rebuild it, but that script writes its output back to `data/stan_model.RDS`, so running the archive in place silently replaces a deposited file with a machine-specific binary. Running the archive from a scratch copy, as this report does, is the only way to leave the deposit intact. Once the model is recompiled, `meta_analysis.R` fails again inside `0_helpers.R`, where `fct_reorder()` is called on a character vector whose ordering variable is `NA` for the predictive-interval row; forcats 1.0.0 errors on that, where earlier versions did not.

The four simulation scripts fail for two reasons that have nothing to do with the analysis either. `Figure2ab.R` and `Figure5.R` call `write_rds(path =)` and `read_rds(path =)`, and the `path` argument was removed in readr 2.0.0. `Figure3a.R` and `Figure3b.R` print the finished plot object to the console before saving it, which sends it to the default `pdf()` device, which cannot resolve the custom font `ETBembo-RomanLF` and raises "invalid font type." The font is missing on any machine but the authors', so the bare print is fatal everywhere. The DeclareDesign declaration fails on its first line of arithmetic, computing `delta` from `pi_star` one line before `pi_star` is defined, and then on the pre-1.0 DeclareDesign API throughout.

All of it is resolvable, and none of the fixes touches an estimator, a specification or a sample definition. With them applied to a scratch copy, the archive's own `meta_analysis.R` reproduces the published Table 4 in 84 of its 91 cells.

1.2 Does the maintained rewrite reproduce the paper?

It agrees with the deposited archive wherever the two compute the same quantity, and with the paper in 196 of 217 verifiable claims.

The comparison against the archive is the sharper of the two. `ground_truth/extract_archive_values.R` runs the deposit from a scratch copy, with only the recompiled Stan model, the repaired `fct_reorder` call and the two `readr` argument names, and writes what its scripts produce into `ground_truth/`. The two pipelines compute the same quantity in 184 places: all of Table 4, the in-text sentences that restate it, all of Table H.8, the study counts behind Tables G.1 to G.7, the two indifference curves behind Figure 2 and the three region counts behind Figure 5. The largest absolute difference across all of them is 0. Any distance to the published numbers is therefore a property of the software this runs on, not of the rewrite.

The 21 claims that do not match the paper fall into three groups, and none of them is a defect in the rewrite.

13 are places where the article's prose disagrees with the article's own tables, and in every one of them the pipeline agrees with the table. The clearest is the turnout predictive interval: the text says the 95% interval "is 45 points wide," while Table 4 prints [-0.12, 0.26] for that interval, which is 38 points wide, and the code returns 37.4. The same pattern recurs for the vote-buying 95% predictive interval (text -27 to +10, table -25 to +8), the racial-prejudice 95% predictive interval (text -16 to +16, table -5 to +12), the width of the religious-prejudice credible interval (text 13 points, table 11), and the sexual-orientation underreporting estimate (text +2 points, table -0.01). The last of the group is the RMSE indifference threshold at N = 1,000: the article says a bias above 5.5 points prefers the list experiment, and the curve its own Figure 2a plots crosses at 5.57, so a bias of 5.55 points would not in fact prefer it. None of these is a reproduction failure. The pipeline and the tables agree with each other, and the prose agrees with neither. They are listed sentence by sentence, each with the corrected wording, in `errata.qmd`, which renders to `blair_coppock_moor_2020_errata.pdf` at the repository root.

6 are cells of Table 4 where the archive and the rewrite return the same posterior summary and the article prints a different one: four predictive-interval bounds, one standard error and one credible-interval bound. All six are in the second decimal place, and five of them are in the two subsets with the fewest studies, where the posterior is least well determined. Stan and its compiler have both moved since 2020 and the deposit pins neither, so the article's draws cannot be recovered.

2 are the counts of studies in the second and third regions of Figure 5. The article reports 98, 107 and 59 studies; the deposited `Figure5.R` and the rewrite both return 98, 106 and 60. One study, Luks (2010), has an estimated bias of exactly 0.2000, which sits one grid step of 0.0001 above the power curve at its sample size and so falls above the curve rather than between the curves. Nothing in that computation draws at random, so the difference is not sampling noise, and what moved between 2020 and now has not been established. Recording it as unresolved is more honest than attributing it.

29 recorded quantities carry no verdict, for three different reasons. 15 are claims about shape, sign or spread that name a rough magnitude and no precision to match it at: "typically smaller than 10 percentage points", "approximately five points", "close to 1", "a considerable range of bias from very large to none at all". Each of those rows carries the computed distribution beside the sentence and no verdict, because a threshold the sentence never states is not ours to invent. Six of the 12 remaining are quantities the article describes without printing: the four Deming regression coefficients behind Figure F.10 and the two thresholds Figure 3b plots. The 2 the deposit cannot support are the counts of list experiments that pass the Blair and Imai (2012) design effects test, which needs an item-count distribution the deposited study-level file does not carry. The last six are published floats that carry no computed quantity at all, or that the deposit contains no script and no data for: Figure 1 and Figure A.1 are drawn from respondent-level surveys the deposit does not include, and Figure D.2 has two panels drawn from data the deposit does carry and two, the variance of the control item count and its covariance with the direct question, drawn from data it does not.

One further observation belongs here rather than in the errata, because it is not a token a reader would quote. Table 4's note says that studies for other categories are not broken out because they "all have fewer than three studies." That is true of the four literatures the paper names, where the only cell left out holds two studies. It is not true of the data as a whole: the residual topic categories the meta-analysis carries hold up to 104 studies each and are left out for a different reason than the note gives. No estimate changes.

2 Paper overview

Citation: Blair, G., Coppock, A. and Moor, M. (2020). "When to Worry about Sensitivity Bias: A Social Reference Theory and Evidence from 30 Years of List Experiments." *American Political Science Review*, 114(4), 1297-1315.

Research question: How much are survey responses on sensitive topics distorted by respondents' concern about how a social referent will judge them, and when is a list experiment worth its cost?

Design: Three pieces. A social reference theory says sensitivity bias requires a referent, a perceived preference of that referent, a belief that the answer will reach the referent, and a cost. A bias-variance analysis shows the list experiment buys unbiasedness at a very high price

in precision: at 2,000 subjects and a prevalence rate of 0.5, its variance is roughly fourteen times the direct question's, so a researcher would need about 28,000 subjects to match a 2,000 subject direct question. A meta-analysis of 264 list experiments from 92 papers, each paired with a direct question, then estimates the actual sensitivity bias by domain with a Bayesian random effects model fitted in Stan.

Main finding: Biases are typically under 10 percentage points and in several domains indistinguishable from zero. The exception is support for authoritarian regimes, where overreporting averages +14 points, and the predictive intervals are wide enough everywhere that a domain-level average is a poor guide to any particular study.

Data: the deposit contains no individual-level microdata. `estimates_df.RDS` is a curated frame of 264 study-level estimates, their standard errors, sample sizes, survey modes and predicted polarity, assembled by the authors from published and unpublished sources.

3 Original archive reproducibility

Table 1: Status of each script in the deposited archive under R 4.6.0.

Script	Status
1_meta_analysis/0_helpers.R	Sourced by the meta-analysis scripts; carries the fct_reorder failure
1_meta_analysis/compile_stan_model.R	Runs, but writes its output over the deposited data/stan_model.RDS
1_meta_analysis/meta_analysis.R	Fails: unusable Stan binary, then fct_reorder on an NA, then no figures/ or tables/ directory
1_meta_analysis/meta_analysis_rma.R	Runs clean; writes no file and corresponds to no published quantity
1_meta_analysis/mode_appendix_table.R	Fails: no tables/ directory in the deposit
1_meta_analysis/plot_direct_v_list.R	Fails: no figures/ directory in the deposit; geom_errorbarh deprecation warning
1_meta_analysis/table_of_data.R	Fails: no tables/ directory in the deposit
2_simulations/theory_list_power_function.R	Sourced by the simulation scripts; runs clean
2_simulations/Figure2ab.R	Fails: write_rds(path =) removed in readr 2.0.0; no figures/ directory
2_simulations/Figure3a.R	Fails: bare print of the plot to a device that cannot resolve the font
2_simulations/Figure3b.R	Fails: same bare print
2_simulations/Figure5.R	Fails: read_rds(path =) removed; also needs Figure2ab.R to have run
3_declaredesign/list_experiment_declaration.R	Fails: delta computed before pi_star; pre-1.0 DeclareDesign API throughout

Four further observations about the deposit itself.

The archive's `README.txt` describes a script named `theory_group_differences_sensitivity_bias.R` that is not in the deposit; the file that produces the figure it describes is `Figure3a.R`. The README also lists the four simulation scripts by filename with no description at all.

`compile_stan_model.R` writing to `data/stan_model.RDS` is the more serious problem, because the output and one of the deposited inputs are the same file. A reader who follows the archive's own instructions ends up with a working copy of the deposit that no longer matches the bytes Dataverse serves, and nothing warns them. The maintained rewrite recompiles the model in memory and never writes a compiled binary at all.

`Figure2ab.R` walks a 3,410,341 row parameter grid one row at a time with `map2_df`, which takes about a quarter of an hour, and then writes the resulting 246 MB frame to a 253 MB file in the archive's root directory. The function it calls, `list_power`, is already fully vectorized, so a single call over the whole grid returns the identical frame in under a second: on a random sample of 4,000 grid points the two routes agree exactly, cell for cell. Vectorizing is a performance property and not an analytical one, and the rewrite takes the fast path.

The zip carries a `__MACOSX` resource-fork tree, three `.DS_Store` files and an `.Rapp.history` alongside the 13 scripts and 2 data files. `download_original.R` discards the resource forks on unpacking and leaves everything else as deposited. Running the archive adds files as well: the two simulation scripts that print a plot before saving it leave an `Rplots.pdf` behind, and `compile_stan_model.R` overwrites `data/stan_model.RDS`, so a working copy of the deposit stops matching Dataverse the first time it is used. `download_original.R` refuses to proceed if `original/` holds anything the manifest does not list, which is why the deposit is kept as the downloaded zip and unpacked somewhere else.

4 Ground truth summary

Table 2: Ground truth claims by source and outcome.

Table or figure	Status	N
Appendix F	Does not match	1
Appendix F	Matches	3
Appendix F	Not stated in the paper	3
Figure 1	Not stated in the paper	1
Figure 2a	Does not match	1
Figure 2a	Matches	1
Figure 2b	Matches	2
Figure 3a	Matches	2
Figure 3b	Not stated in the paper	2
Figure 4a	Matches	5
Figure 5	Does not match	2
Figure 5	Matches	2
Figure A.1	Not stated in the paper	1
Figure D.2	Not stated in the paper	1

Figure E.3	Matches	1
Figure E.4	Matches	1
Figure E.5	Matches	1
Figure E.6	Matches	1
Figure E.7	Matches	1
Figure E.8	Matches	1
Figure E.9	Matches	1
Figure F.10	Not stated in the paper	4
Table 1	Not stated in the paper	1
Table 2	Not stated in the paper	1
Table 3	Not stated in the paper	1
Table 4	Does not match	6
Table 4	Matches	84
Table 4	Not stated in the paper	1
Table G.1	Matches	5
Table G.2	Matches	2
Table G.3	Matches	2
Table G.4	Matches	2
Table G.5	Matches	2
Table G.6	Matches	2
Table G.7	Matches	2
Table H.8	Matches	36
Tables G.1 to G.7	Matches	1
Text, abstract	Not stated in the paper	2
Text, authoritarian regimes	Does not match	1
Text, authoritarian regimes	Matches	9
Text, bias-variance trade-off	Matches	6
Text, discussion	Matches	1
Text, limitations	Not stated in the paper	4
Text, meta-analysis design	Matches	2
Text, racial prejudice	Does not match	3
Text, racial prejudice	Matches	6
Text, racial prejudice	Not stated in the paper	2
Text, religious prejudice	Does not match	1
Text, religious prejudice	Matches	2
Text, religious prejudice	Not stated in the paper	1
Text, sexual orientation	Does not match	1
Text, sexual orientation	Matches	2
Text, sexual orientation	Not stated in the paper	2
Text, summary	Not stated in the paper	1
Text, turnout	Does not match	1
Text, turnout	Matches	4
Text, vote buying	Does not match	4
Text, vote buying	Matches	4
Text, vote buying	Not stated in the paper	1

Every published number enters through `ground_truth/published_claims.csv`, the extraction: 313 numeric claims made by the article and its supplementary materials, each classified by hand as pipeline-derived, definitional, structural, transcribed from another paper, or descriptive. The 234 classified pipeline or descriptive are checked twice, once by a row in the ground truth

and once by a block in `maintained/in_text_claims.R` that reaches the same number out of the same output files by its own path. `build_ground_truth.R` runs that file non-interactively, counts the claims it printed against the extraction, and stops if the two instruments land on different numbers. Running it rather than scanning it for markers is the point: a block that errors on its first line, or that ends in a bare expression and so prints nothing under `source()`, passes a textual check while verifying nothing.

Appendix G is covered cell by cell rather than by a summary row. Its seven tables print 717 numbers across 239 study rows, read off the published pages positionally by `ground_truth/extract_published_values.R` and compared in the order the pages print them; 717 of them reproduce. Appendix E's seven figures print no estimate on their faces, so what is checked there is how many study rows each one draws, also counted off the page.

`value_paper` in `ground_truth/blair_coppock_moor_2020_ground_truth.csv` was filled only by reading the published article and its supplementary materials, and the three multi-column tables were read off rendered pages rather than a text extraction so that no cell could be mapped to the wrong column. Where the paper does not state a quantity the field is blank and `match` is `NA`. `value_rewrite` is read back out of `maintained/output/` by the build script and `value_script` out of the archive run, neither of them transcribed, and no published number is an input to any computation anywhere in `maintained/`. Nor is a published number used to select a row: the one study-level spot check is on Corstange (2017), which is the single vote-buying study with one row in Table G.1 and so can be located by name.

Claims the article prints as values are matched to the last digit it prints, by comparing formatted strings rather than by a numeric tolerance, because a double does not record how many decimals a page used. A handful are stated as one-sided bounds instead ("the bias must be greater than 5.5 points", "differences must exceed 20 percentage points"), and those are tested as bounds, in the direction the sentence asserts. A third group names a rough magnitude and no precision at all ("approximately five points", "close to 1"), and those carry the computed quantity beside the article's figure and no verdict.

Table 3: Every claim that does not match, and where the difference comes from.

Source	Claim	Paper	Pipeline	Locus
Table 4	turnout overreporting 95% PPI upper	0.26	0.255	environment
Table 4	racial prejudice underreporting 95% PPI upper	0.12	0.128	environment
Table 4	sexual orientation overreporting mu SE	0.09	0.079	environment
Table 4	sexual orientation overreporting 95% CI lower	-0.19	-0.173	environment
Table 4	sexual orientation overreporting 95% PPI lower	-0.31	-0.301	environment
Table 4	sexual orientation overreporting 95% PPI upper	0.27	0.264	environment
Figure 2a	bias at which list and direct RMSE are equal, N = 1000, points	5.5	5.570	paper_internal

Figure 5	studies between the two curves	107	106.000	unresolved
Figure 5	studies above both curves	59	60.000	unresolved
Text, vote buying	95% credible interval upper, points	-3	-3.690	paper_internal
Text, vote buying	50% predictive interval upper, points	-2	-2.674	paper_internal
Text, vote buying	95% predictive interval lower, points	-27	-24.852	paper_internal
Text, vote buying	95% predictive interval upper, points	10	8.148	paper_internal
Text, racial prejudice	95% credible interval lower, points	-3	-2.421	paper_internal
Text, racial prejudice	95% predictive interval lower, points	-16	-5.199	paper_internal
Text, racial prejudice	95% predictive interval upper, points	16	12.777	paper_internal
Text, sexual orientation	underreporting meta-analysis estimate, points	2	-0.966	paper_internal
Text, authoritarian regimes	overreporting 50% predictive interval lower, points	4	4.516	paper_internal
Appendix F	all underreporting: meta-analysis estimate, points	-3	-3.517	paper_internal
Text, turnout	width of the 95% predictive interval, points	45	37.368	paper_internal
Text, religious prejudice	width of the 95% credible interval, points	13	11.391	paper_internal

5 Maintained rewrite

5.1 Architecture

Table 4: The maintained rewrite, one script per output.

Script	Output
helpers.R	Packages, the Stan helpers, the plotting helpers, the table formatters
figure_4_meta_analysis_coefplots.R	32 figure files (Figure 4a-h and Figures E.3-E.9), figure_4_meta_estimates.csv, meta_analysis_summaries.rds
table_4_meta_summary.R	table_4_meta_summary_cat.csv, table_4_meta_summary_overall.csv
figure_2ab_theory_power_rmse.R	figure_2ab_theory_power_rmse.pdf/.png, figure_2ab_indifference_curves.csv
figure_3a_group_differences_power.R	figure_3a_group_differences_power.pdf/.png, figure_3a_power_curve.csv
figure_3b_diff_in_sensitivity_power.R	figure_3b_diff_in_sensitivity_power.pdf/.png, figure_3b_power_curve.csv
figure_5_bias_variance_overlay.R	figure_5_bias_variance_overlay.pdf/.png, figure_5_region_counts.csv
figure_f10_direct_vs_list.R	figure_f10_direct_vs_list.pdf/.png, figure_f10_deming_coefficients.csv

table_g_study_estimates.R	table_g_study_estimates.csv (Tables G.1-G.7 in one tidy file)
table_h8_mode.R	table_h8_mode.csv
text_theory_variance.R	text_theory_variance.csv
text_power_thresholds.R	text_power_thresholds.csv
text_summary_stats.R	text_summary_stats.csv
declaredesign_list_experiment.R	declaredesign_simulation_summary.csv
in_text_claims.R	Printed only: every claimed number beside the sentence that makes it

Three structural points are worth naming.

Every figure script writes a CSV of the values it plots. A PDF differs on every run because it records the time it was written, so a figure file cannot be diffed; the CSV beside it can. For Figure 4 alone that means 389 rows of plotted estimates, standard errors, credible intervals and predictive intervals, and the same holds for the two indifference curves behind Figure 2, the two power curves behind Figure 3, the three region counts behind Figure 5 and the four Deming coefficients behind Figure F.10.

The ten Stan fits run once, in `figure_4_meta_analysis_coefplots.R`, which saves their posterior summaries to `output/meta_analysis_summaries.rds`; `table_4_meta_summary.R` reads that file rather than refitting. One script per output means the figure and its table are written by different scripts, and passing the fits through a saved object is what stops the two from ever reporting different draws. The archive keeps both in one script and gets the same guarantee a different way.

Three scripts exist only to make in-text numbers traceable. `text_theory_variance.R` computes the standard errors quoted in the theory section, which the archive works out only in a commented block at the foot of `theory_list_power_function.R`, and that block uses a sensitivity bias of 0.05 and a control item covariance of 0.05 rather than the 0.1 and 0.025 the published sentence states. `text_power_thresholds.R` reads the three simulation curves and prints the sample-size thresholds the design section quotes. `text_summary_stats.R` gives the study counts and the extreme study estimates named in the results. Where the archive prints a number to the console and saves nothing, as `Figure5.R` does with its region counts, the rewrite writes it to a CSV instead.

5.2 Deprecated patterns replaced

Table 5: Deprecated and removed patterns, and what replaced them.

Original pattern	Replacement
‘%>%‘	‘|>‘
‘readRDS()‘ / ‘saveRDS()‘	‘read_rds()‘ / ‘write_rds()‘
‘write_rds(path =)‘, ‘read_rds(path =)‘	positional first argument, ‘file =‘ on write

‘gather()‘	‘pivot_longer()‘
‘do(predict(...))‘, ‘do(tidy(...))‘	‘reframe()‘
‘geom_errorbarh(height = 0)‘	‘geom_linerange()‘
‘fct_reorder()‘ on a character vector with NA	sort the NA row to ‘-Inf‘ before reordering
‘geom_smooth(stat = ”smooth”)‘	‘geom_smooth(method = ”loess”)‘
‘xtable‘ + ‘print.xtable(file =)‘	‘write_csv()‘ to ‘maintained/output/‘
‘declare_population()‘, ‘declare_step()‘	‘declare_model()‘ with the step folded in
‘declare_potential_outcomes()‘	‘potential_outcomes()‘ plus ‘declare_measurement()‘
‘declare_estimand()‘	‘declare_inquiry()‘
‘declare_assignment(prob = 0.5)‘	‘declare_assignment(Z = complete_ra(N, prob = 0.5))‘
‘declare_estimator(model =)‘	‘declare_estimator(.method =)‘
‘tidy_estimator()‘	‘label_estimator()‘
‘text = element_text(family = ”ETBembo-RomanLF”)‘	removed; the font is not distributable and the figures do not need it
‘map2_df()‘ over a 3.4 million row grid	one vectorized call to the same function
pre-compiled ‘stan_model.RDS‘	recompiled in memory from the deposited ‘.stan‘ source

Two of these deserve more than a table row.

`tidy_estimator()`. The archive's DeclareDesign declaration has three estimators: the direct question, the list experiment, and the difference between them with a bootstrapped standard error. The third is wired in with `tidy_estimator()`, which DeclareDesign removed at version 1.0. Its direct successor is `label_estimator()`, the estimator declares in one line, and the whole simulation with 500 designs and 250 bootstrap resamples each runs in about ninety seconds. Across those 500 simulated designs the direct-minus-list estimator recovers the true sensitivity bias of 0.1 with a simulation bias of -0.0003 and power of 0.67. The four control items are summed inside the expression that draws them, because fabricatr 2.0 splits a matrix-valued column into `control_item.1` through `control_item.4` as it stores it, so a later `control_item[, 1]` finds nothing. No number from this script appears in the published paper; the declaration is a worked example of the design the paper analyses.

The custom font. Every figure in the archive sets `family = "ETBembo-RomanLF"`, a commercial typeface that is not in the deposit. On the authors' machine this is invisible; anywhere else it is either a silent fallback or, when the plot reaches the base `pdf()` device, a hard error reading “invalid font type”. The rewrite removes the font calls entirely, and the figures render on any machine. They are otherwise the published figures, with one cosmetic omission: the archive draws the two labels inside Figure 5 with a leading arrow glyph taken from the Adobe Symbol font, and the rewrite prints the labels without it.

5.3 What the rewrite does not carry

`meta_analysis_rma.R` refits all ten subsets with `metafor::rma.uni(method = "EB")` and then stops: it writes no file, and no quantity it computes appears in the article or the supplementary materials. It runs clean, and it was run for this report to confirm that. It is not reproduced in `maintained/` because there is nothing published for it to reproduce.

The two bug fixes the rewrite makes, the `fct_reorder` repair and the `pi_star` ordering in the design declaration, are both fixes to code that crashes rather than to code that computes the wrong answer, and neither moves a published value. There is no erratum to report.

6 Figure verification

Table 6: Published figures and the maintained scripts that produce them.

Figure	Rewrite output	Numbers checked against the paper
Figure 2a, 2b	figure_2ab_theory_power_rmse	RMSE and power thresholds at N = 1,000 and N = 2,000
Figure 3a	figure_3a_group_differences_power	Detectable prevalence difference at N = 2,000 and N = 3,000
Figure 3b	figure_3b_diff_in_sensitivity_power	None stated in the text
Figure 4a-h	figure_4a_* through figure_4h_*, plus _sm small multiples	Study-level range and the walked-through Kiewiet de Jonge estimate
Figure 5	figure_5_bias_variance_overlay	Counts of studies in the three regions
Figure F.10	figure_f10_direct_vs_list	None printed; slopes described only as close to 1

Figure 4 is the paper's central display, eight panels of study-level sensitivity bias estimates with the meta-analytic posterior and its predictive intervals underneath. The rewrite reproduces all eight, and the same code path produces Figures E.3 through E.9 in the supplementary materials.

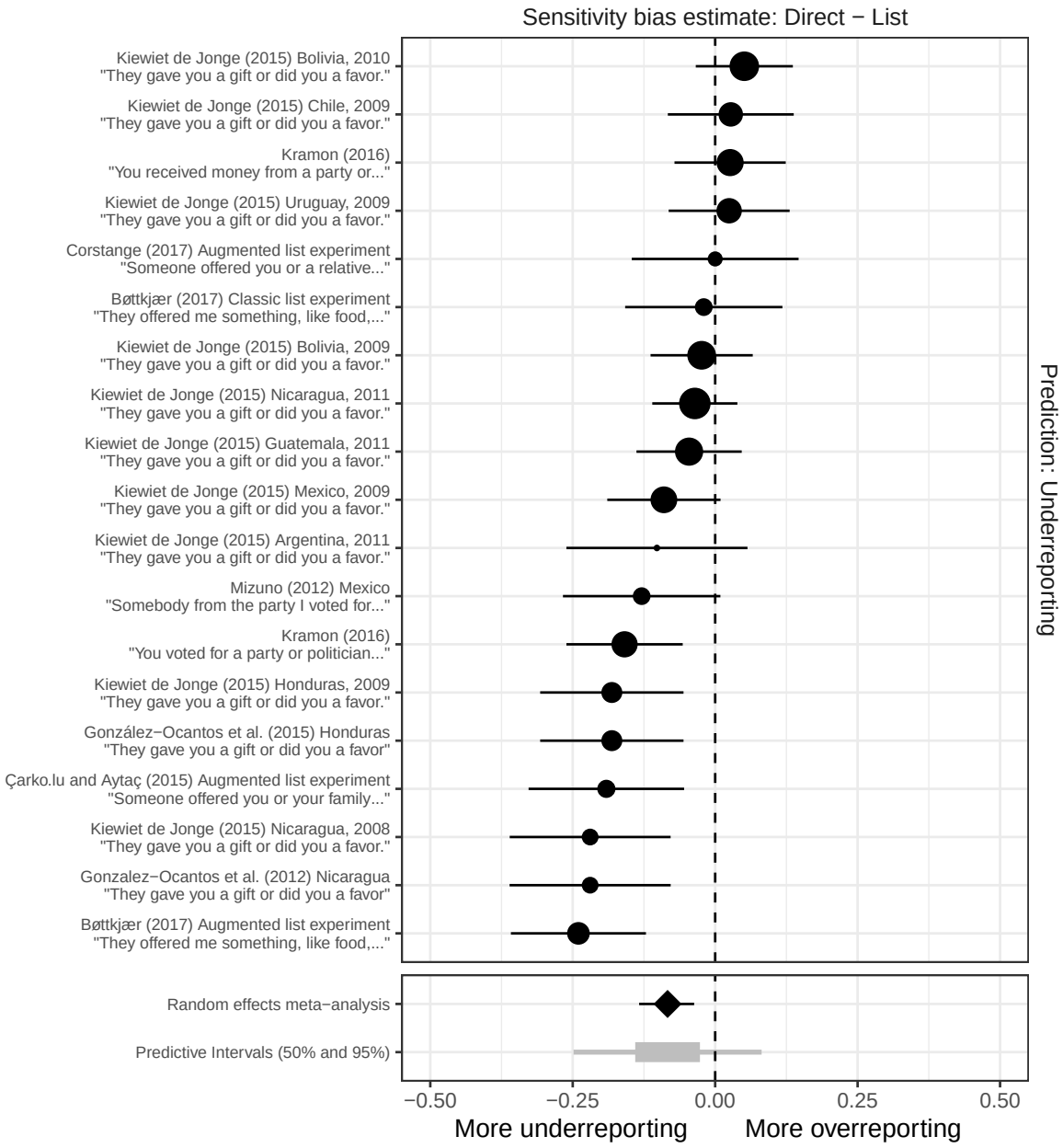


Figure 1: Figure 4a as reproduced: vote buying, underreporting predicted.

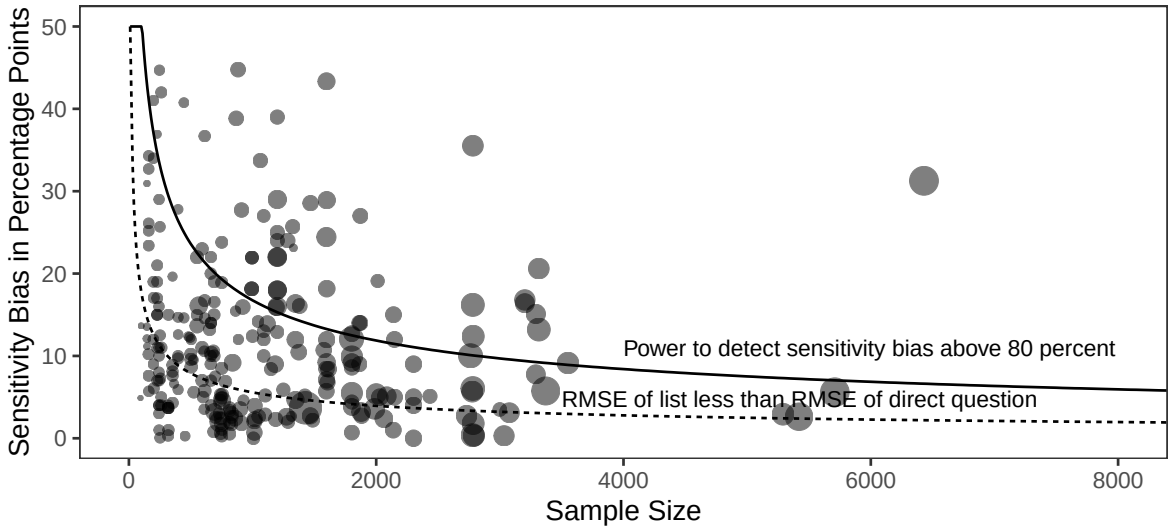


Figure 2: Figure 5 as reproduced: where the 264 list experiments fall relative to the two indifference curves.

7 Rewrite verification

Table 7: Table 4, by subset: how many of the nine printed quantities the rewrite reproduces.

Domain	Matching	Not matching
vote buying underreporting	9	0
all results underreporting	9	0
all results overreporting	9	0
turnout overreporting	8	1
racial prejudice underreporting	8	1
religious prejudice underreporting	9	0
sexual orientation overreporting	5	4
sexual orientation underreporting	9	0
authoritarian overreporting	9	0
authoritarian underreporting	9	0
largest domain by polarity cell Table 4 does not break out, in studies	NA	NA

The rewrite and the archive agree on every one of these cells, including the six that disagree with the paper. That is the useful comparison: a rewrite defect would show up as the rewrite disagreeing with the archive, while software drift shows up as both disagreeing with the paper in the same direction by the same amount. All six sit in the second decimal place of a posterior summary, and five of them are in the two subsets with the fewest studies.

The Stan model draws 10,000 iterations on four chains at a fixed seed of 500, set once before the first fit. Running the pipeline twice returns byte-identical CSV, PNG and RDS output, including the saved posterior summaries, so the remaining distance to the published values is attributable to the versions of rstan and R rather than to sampling noise within a version.

8 R environment

Table 8: Versions this report was produced under.

Component	Version
R	4.6.1
rstan	2.32.7
tidyverse	2.0.0
tidybayes	3.0.7
DeclareDesign	2.0.0
deming	1.4.1
janitor	2.2.1
patchwork	1.3.2

The archive was built under R 3.6.3 and run on 8 May 2020.